



Human Computer Interaction

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Mobile User Interfaces

Mobile User Interfaces Topics



Design challenges in mobile UIs

- specifically modern smartphones

Design patterns & guidelines

Desktop vs. Mobile

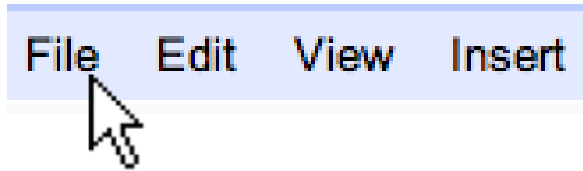


Small Screen



“Fat Finger”

large contact surface of a fingertip (compared to a mouse pointer or stylus)



Lorem Ipsum is simply dummy text of the printing industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets of Lorem Ipsum patterns, and more recently with desktop publishing software like Aldus PageMaker which included versions of Lorem Ipsum.



Text Input

Slower and more error-prone
full-sized keyboard:

- average 30-40 wpm
- fast 50-80 wpm
- peak 120 or more



phone keyboard:

- fast typists probably manage **half their speed** on a fullsize

Context



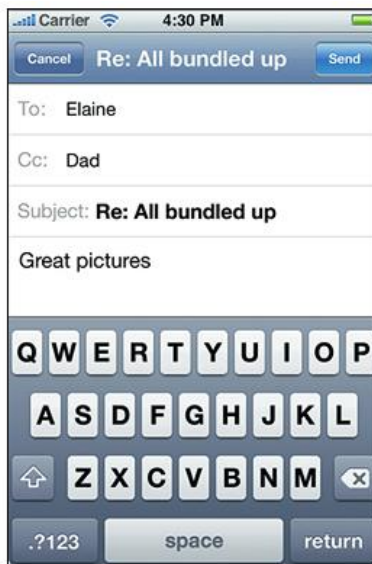
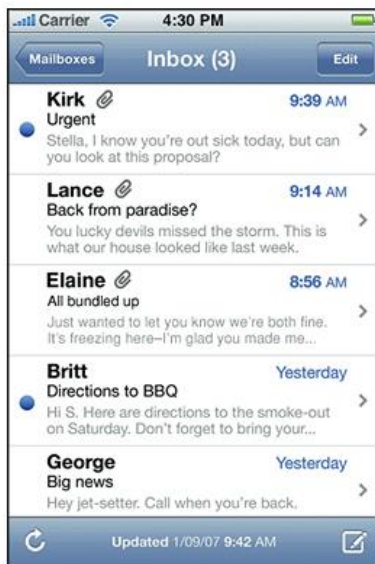
Other Issues in Mobile

Power & battery life

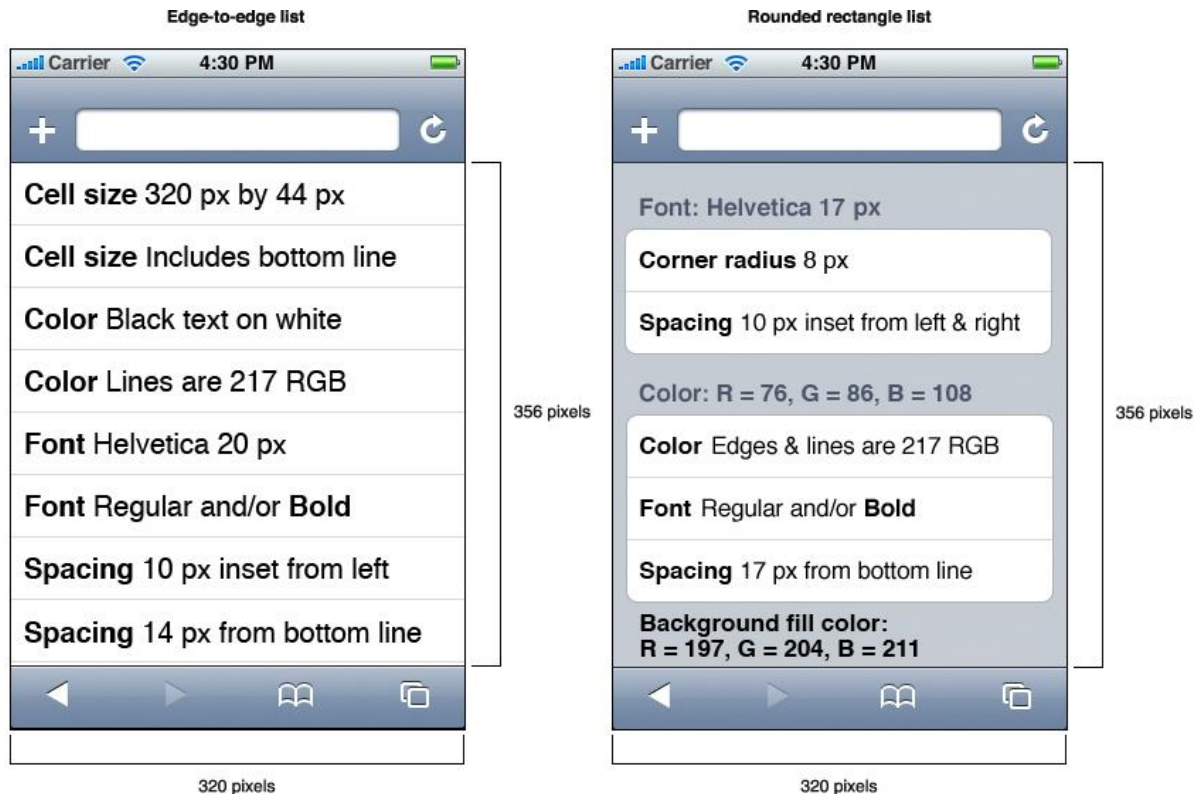
Network latency, bandwidth, inconsistency

CPU speed

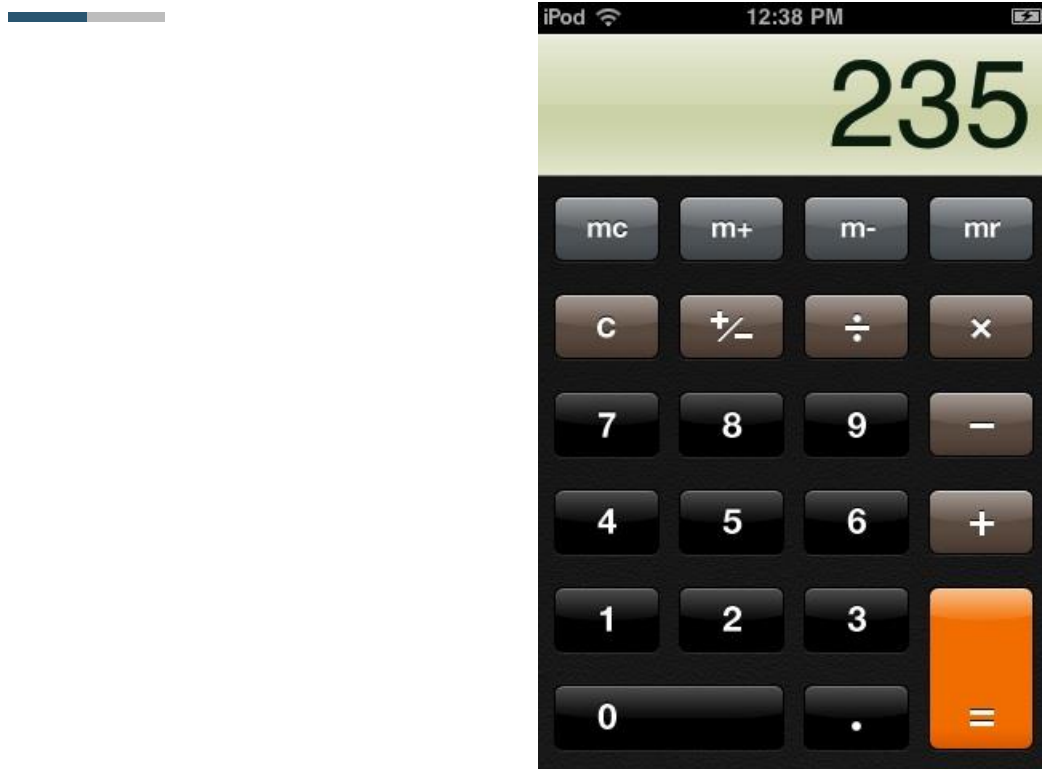
Distinct Screens



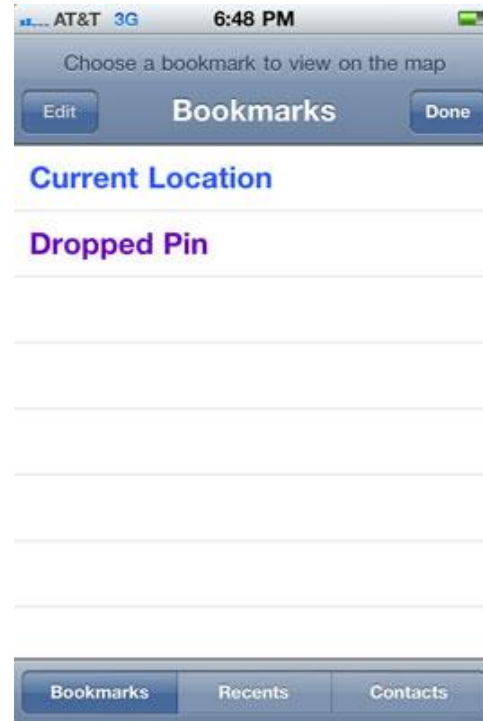
Scrolling Lists



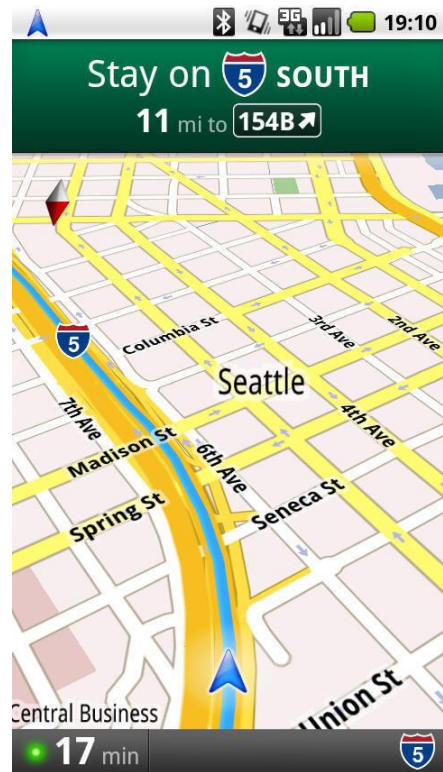
Finger-Sized Targets



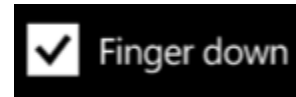
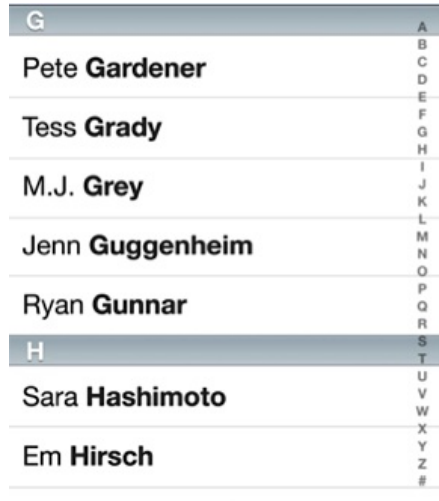
Minimize Text Input



Simplify, Simplify, Simplify!



Mobile Widgets



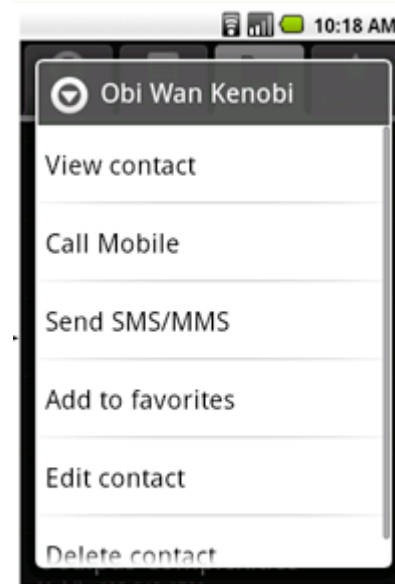
Many Kinds of Menus



Options icon menu



Options expanded menu



Context Menu

Touch Gestures



Summary

Mobile UI design faces new challenges

- small screens
- fat fingers
- poor text entry

Simplify

- Follow design patterns
- Use touch gestures where possible



Ubiquitous Computing

What is Ubiquitous Computing?



The method of **enhancing computing** use by making **many devices** (services) **available** throughout the **physical environment** but making them effectively **invisible** to the **user**.
(Mark Weiser)

Mark Weiser:

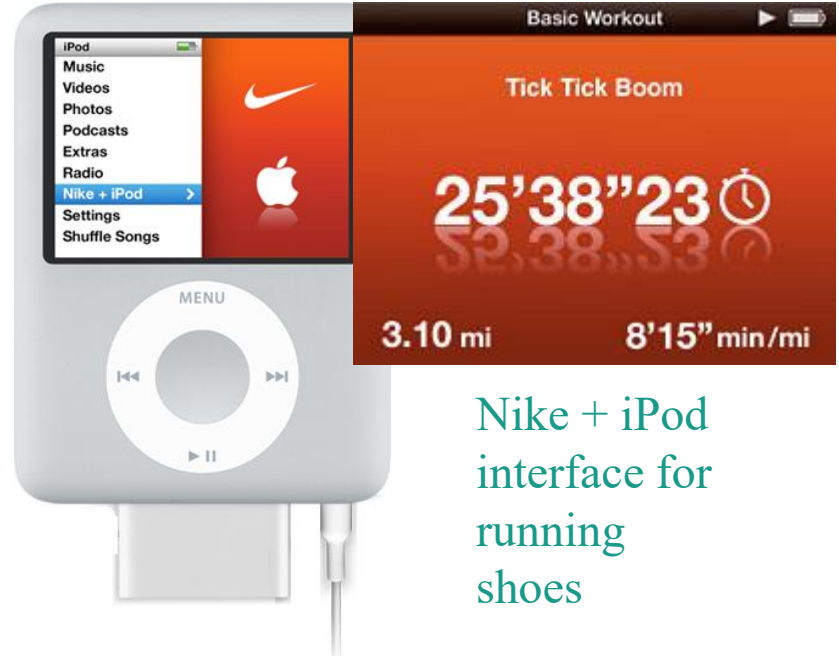
The father of Ubiquitous Computing

- Researcher in the Computer Science Lab at Xerox's PARC (Palo Alto Research Center)
- first articulated the idea of ubiquitous computing in 1988
- has called UC "...highest ideal is to make a **computer so imbedded, so fitting, so natural**, that we use it **without** even **thinking** about it."

Ubiquitous Computing

Move beyond desktop machine

Computing is embedded everywhere in the environment



Nike + iPod
interface for
running
shoes

Ubiquitous Computing Examples

Internet Kitchen

Internet refrigerator and cooking appliance

Download recipes from web directly to device



Automatic hand
soap dispenser



Computing Everywhere



Ubiquitous means:

- Present **everywhere**
- **Simultaneously** encountered in numerous **different instances**
- Computers become a useful but **invisible** force, **assisting** the user in meeting his **needs without** getting **lost in the way**

What is Ubiquitous Computing? (Cont.)

- Computing everywhere

- Many **embedded, wearable, handheld** devices **communicate transparently** to provide different **services** to the **users**.
- Devices mostly have **low power** and **short-range wireless** communication capabilities.
- Devices **utilize** multiple **on-board sensors** to **gather information** about surrounding environments.

Context-Awareness

- **Computers** will be able to **understand enough** of a user's current **situation** to **offer services, resources, or information relevant** to the particular context.
- The **attributes** of **context** to a particular situation vary widely, and may include the **user's location, current role** and other elements.
- The application of context may include any combination of these elements.

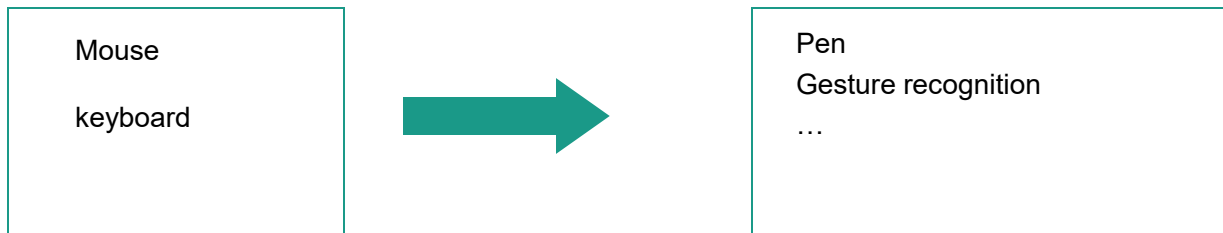
Challenges and Requirements

- Hardware
- Applications
- **User Interfaces**
- Networking
- Mobility
- Scalability
- Reliability
- Interoperability
- Resource Discovery
- Privacy and Security

Users Interface

The multitude of different Ubicomp devices with **their different sizes of displays and interaction capabilities** represents another challenge.

Rather than **force** the user to search out and **find the computer's interface**, ubiquitous computing **suggests** that the **interface itself** can **take** on the **responsibility** of **locating** and **serving** the user.



Scales of devices



Weiser proposed

- Inch
- Foot
- Yard

Implications for device size as well as relationship to people

Device scales



Inch

- PDAs
- Voice Recorders
- smart phones

Individuals **own many** of them and they can **all communicate** with each other and environment.

Device scales

Foot

- notebooks
- tablets
- digital paper



Individual **owns** several but **not assumed** to be **always** with them.

Device scales

Yard

- electronic whiteboards
- plasma displays
- smart bulletin boards



Buildings or institutions own them and lots of people share them.

HCI Challenges

- **Defining** the appropriate physical interaction experience;
- **Discovering** general application features;
- **Theories** for **designing** and **evaluating** the human experience within ubicomp.

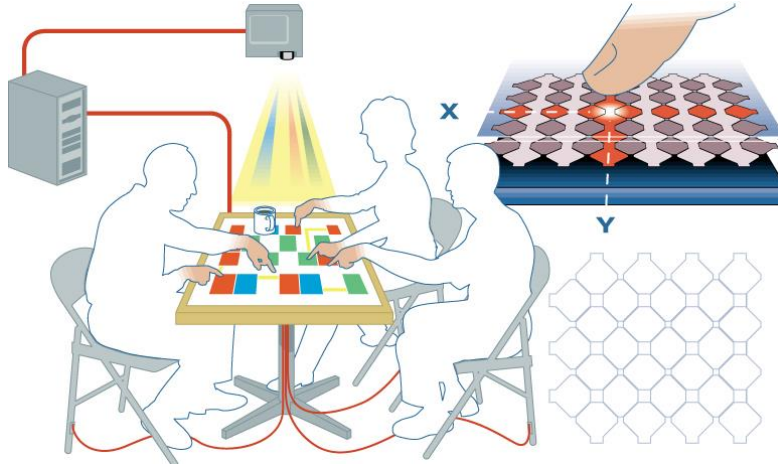
Defining the Interaction Experience

Implicit input

- Sensor-based input
- Extends traditional explicit input (e.g., keyboard and mouse)
- Towards “awareness”
- Use of recognition technologies (simple sensors such as **RFID**, **tilt sensors**, infrared range finders, etc...)

Invisibility of computing, can start when we are able to **determine** an **individual's identity**, **location**, **affect** or **activity** through her/his mere **presence** and natural **interactions** within an **environment**.

Different Inputs



Capacitive sensing on a table

Proximity range sensor:

Infrared (IR) receiver

IR emitter (below receiver to right)

Touch sensitivity:

Screen bezel

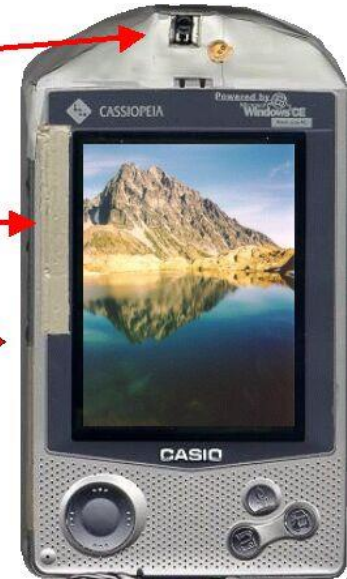
On sides & back of device

Tilt sensor:

Inside device, in plane of the display

2-axis linear accelerometer

Sensors on a PDA



Multi-scale and distributed output

Screens of many sizes

- (very) small



- (very) large



Distributed in space, but coordinated

Application Themes

Context-aware computing

- Sensed phenomena facilitate easier interaction (Ex: **GPS** outside, but what about inside?)

Automated capture and access

- Live experiences stored for future access

Toward continuous interaction

- Everyday activities have **no clear begin-end conditions**

New Opportunities for Theory



Knowledge in the world

- Ubicomp places more **emphasis** on the **physical world**

Activity theory

- Goals and actions fluidly adjust to physical state of world

Ethnography

- Deep descriptive understanding of activities in context

Evaluation Challenges

How can we adapt other HCI techniques to apply to ubicomp settings?

- Ubicomp activities **not** so **task-centric**
- Metric of success could be very different (**playfulness**, non-distraction versus efficiency)

Sources



User Interface Design and Implementation MIT Open Course
Ware ocw.mit.edu

Human-Computer Interaction, Third Edition, Alan Dix, Janet Finlay,
Gregory D. Abowd, Russell Beale, Pearson.

Thank You